

Numbers and Significant Figures Quiz

Name: Key

Question 1

Give the total number of significant figures and the position of the least significant digit in each quantity"

<u>1.250</u> g	<u>6.022</u> × 10 ²³	0.00 <u>215</u> L	<u>3500</u> km	<u>2.590</u> × 10 ⁻⁷ m
4, 10 ⁻³	4, 10 ⁻³ (10 ²⁰)	3, 10 ⁻⁵	2, 10 ²	4, 10 ⁻³ (10 ⁻¹⁰)

Question 2

Give the solution to each expression with the proper number of significant figures

$x = 23.14 \text{ cm} + 4.105 \text{ cm}$	$x = \frac{0.12 \text{ mol}}{1.53 \text{ L}}$	$x = 94 \mu\text{s} - 8.7 \times 10^{-5} \text{ s}$	$x = \frac{12.4 \text{ g} + 1.94 \text{ g}}{20.4 \text{ cm}^3 - 3.47 \text{ cm}^3}$
27.24 cm	0.078 mol/L	7 · 10 ⁻⁶ s	$\frac{14.34 \text{ g}}{16.93 \text{ cm}^3} = 0.847 \text{ g/cm}^3$

Question 3

Later in this course we will use the following equation: $v_{rms} = \sqrt{\frac{3RT}{M}}$ Find the units of v_{rms} if R has units $\frac{\text{J}}{\text{mol K}}$, T has units K , M has units $\frac{\text{kg}}{\text{mol}}$, and $J \equiv \frac{\text{kg m}^2}{\text{s}^2}$

$$\left(\frac{\frac{\text{J}}{\text{mol K}} \cdot \text{K}}{\frac{\text{kg}}{\text{mol}}} \right)^{1/2} \rightarrow \left(\frac{\text{J}}{\text{kg}} \right)^{1/2} \rightarrow \left(\frac{\frac{\text{kg m}^2}{\text{s}^2}}{\text{kg}} \right)^{1/2} \rightarrow \left(\frac{\text{m}^2}{\text{s}^2} \right)^{1/2} \rightarrow \text{m/s}$$

Question 4

Complete the following table:

Decimal Quantity	Scientific Notation	Prefix Notation
0.0045 m	<u>4.5 · 10⁻³ m</u>	<u>4.5 mm</u>
<u>36,000,000 m</u>	<u>3.6 · 10⁷ m</u>	36 Mm
<u>0.000 000 560 m</u>	5.60 × 10 ⁻⁷ m	<u>560 nm -or- 0.560 μm</u>

Spaces exaggerated for clarity